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Technical proposal (Envelope 1)

Tender NGPA/T/2026/052 · Air Quality and Noise Monitoring Network

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/052, we, Nimbrel Sensing Ltd, submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Priya Natarajan
Commercial Director, Nimbrel Sensing Ltd

Date: 9 December 2026
Reference: NS-NGPA-052

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1. Company and references

Nimbrel Sensing builds dense, low-cost air-quality and noise sensor networks for cities, ports and road operators. Founded in 2017, we have 40 staff and more than 2,000 sensor pods in 14 countries. In the State, installation and first-line maintenance are carried out by our partner Gulf Smart Systems (Al-Noor).

Reference	Scope	Year
City of Westhaven, smart-city air network	120 sensor pods, public dashboard	2022
Northbay Port, fence-line monitoring	18 sensor pods, 6 noise nodes	2024
Coastal Highways Agency, roadside monitoring	40 sensor pods	2025

2. Technical solution

AQ-Pod S5. One compact pod per site, mounted on a pole or a wall, with:

- an optical particle counter for PM1, PM2.5 and PM10; the AQ-Pod S5 is certified under MCERTS for indicative ambient particulate monitors;
- electrochemical sensors for NO2, SO2 and O3, corrected by our machine-learning calibration models;
- wind, temperature and humidity sensors;
- a solar panel and battery, so no mains power or air-conditioning is needed; the pod is rated IP65 and operates from −20 °C to +60 °C;
- 14 days of local storage, with automatic back-fill over 4G.

Data quality. Each pod is first co-located for two weeks with a reference station, and its calibration model is then updated monthly from the network. Typical hourly uncertainty is ±25 % for PM2.5 and ±30 % for NO2. Our networks achieve 95 % data capture.

NoiseNode C2. Class 2 sound level meters to IEC 61672-1, with 1/3-octave spectra and 10-second audio clips of exceedances.

Nimbrel Cloud. Dashboards, automatic quality checks, alarms by push notification, SMS and e-mail within 1 minute of the end of the averaging period, monthly reports in PDF and CSV, a public dashboard in Arabic and English, and an open REST API.

3. Security and data

Hosting. Nimbrel Cloud runs on a public cloud provider in the European Union (Dublin region) and is certified to ISO 27001. Data is encrypted in transit and at rest.

- Authority users sign in with single sign-on (SAML) through the Authority's identity provider.
- Data is kept online for 5 years and archived to the Authority on request.

4. Implementation and timeline

Milestone	Month

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Pods and noise nodes shipped	1
All sites installed, co-location calibration	2
Network live, trial period	3
Acceptance	4

Training. Two online sessions (English) for the Authority's users.

5. Support, warranty and SLA

Warranty. 12 months from acceptance.

Service. We monitor every pod remotely. Faulty pods are replaced from our UK stock within 5 working days; Gulf Smart Systems visits a site within 72 hours when needed. Sensor cartridges are replaced every 12 months, and calibration models are updated every month.

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Annex A. Compliance matrix (Annex 1 of the RFP)

Clause	Requirement	Response	Offered
2.1	Six AQ stations: PM10, PM2.5, NO2, SO2, O3 + meteorology	Comply	AQ-Pod S5 at 6 sites (§2)
2.2	Four noise terminals	Comply	NoiseNode C2 (§2)
3.1	PM analysers certified equivalent (EN 16450)	Partially comply	MCERTS indicative (§2)
3.2	Gas analysers to EN reference methods	Partially comply	Electrochemical sensors with calibration models (§2)
3.3	Noise Class 1, 1/3-octave, audio recording of exceedances	Partially comply	Class 2, audio clips (§2)
3.4	Shelter IP55, 0 to +50 °C, UPS ≥ 2 h	Comply	IP65, solar, no shelter needed (§2)
3.5	7 days local storage, back-fill, backup link	Comply	14 days, 4G (§2)
3.6	≥ 90 % valid hourly data per quarter	Comply	95 % (§2)
4.1	Authority data centre; 10-year retention	Partially comply	Nimbrel Cloud, EU region (§3)
4.2	Validation with audit trail	Comply	§2
4.3	Alarms within 5 min (SMS, e-mail)	Comply	Within 1 minute (§2)
4.4	Monthly / annual permit reports (Excel, PDF)	Comply	PDF and CSV (§2)
4.5	Public dashboard AR/EN + REST API	Comply	§2
4.6	Active Directory sign-in	Comply	SAML single sign-on (§3)
5.1	In operation ≤ 6 months; 30-day trial	Comply	Acceptance month 4 (§4)
5.2	Calibration every 3 months, ISO/IEC 17025 traceable	Comply	Co-location and monthly model updates (§2, §5)
5.3	48 h on-site response; spares in the State	Partially comply	Partner within 72 h (§5)
5.4	24-month warranty; 3-year O&M	Partially comply	12 months (§5)
5.5	Training of 6 HSE staff, AR/EN	Comply	Online sessions (§4)
6.1	≥ 3 similar networks in 7 years	Comply	§1
6.2	Office or service partner in the State	Comply	Gulf Smart Systems (§1)